- Slovakia has five nuclear reactors generating half of its electricity and one more under construction.
- Slovakia's first commercial nuclear power reactor began operating in 1972.
- Government commitment to the future of nuclear energy is strong.

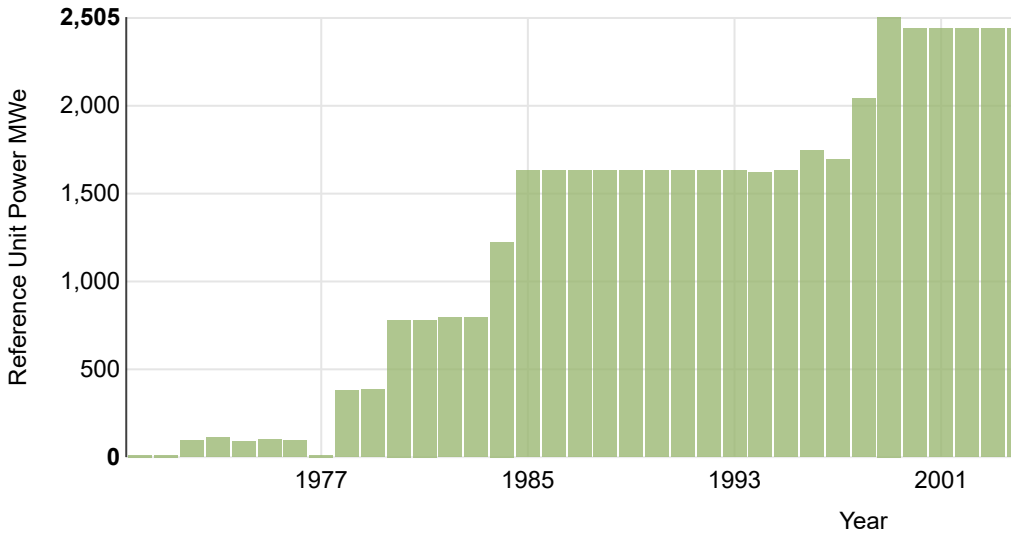
Privacy - Terms

OPERABLE REACTORS
2,302 MWe

REACTORS UNDER CONSTRUCTION
440 MWe

REACTORS SHUTDOWN
912 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 29.9 TWh

Generation mix: nuclear 18.3 TWh (61%); hydro 5.0 TWh (17%); natural gas 2.3 TWh (8%); coal 1.7 TWh (6%); biofuels and waste 1.5 TWh (5%); solar 0.6 TWh; oil 0.4 TWh.

Import/export balance: 3.4 TWh net export (10.6 TWh imports; 14.1 TWh exports)

Total consumption: 21.3 TWh

Per capita consumption: c. 3900 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Electricity consumption in Slovakia has been fairly steady since 1990[a].

Nuclear power industry

Reactors operating in Slovakia

Reactor Name	Model	Reactor Type	Net Capacity (MWe)	Construction Start	First Grid Connection
Bohunice 3	VVER V-213	PWR	466	1976-12	1984-08
Bohunice 4	VVER V-213	PWR	466	1976-12	1985-08
Mochovce 1	VVER V-213	PWR	408	1983-10	1998-07
Mochovce 2	VVER V-213	PWR	408	1983-10	1999-12
Mochovce 3	VVER V-213	PWR	440	1987-01	2023-01

*Construction of Mochovce 3 began in January 1987, but was suspended in March 1993 and resumed in June 2009

Nuclear industry development

Bohunice

In 1958 the Czechoslovak government started building its first nuclear power plant – a gas-cooled heavy water pressure-tube reactor at Bohunice (now in Slovakia). This 110 MWe net Russian-designed Bohunice A1 reactor, built by Skoda, was completed in 1972 and ran until 1977 when it was closed due an accident arising from refuelling[b].

In 1972 construction of the Bohunice V1 plant commenced, with two VVER-440 V-230 reactors supplied by Atomenergoexport of

Russia and Skoda of (then) Czechoslovakia. The first was grid connected in 1978, the second two years later. In 1976 construction started on two V-213 reactors (the V2 plant) built by Skoda. The V2 units commenced operation in 1984 and 1985. All were designed by Atomenergoproekt.

As a precondition for Slovak entry into the European Union (EU) in 2004, the Slovak government committed to closing the Bohunice V1 units 1 and 2 due to perceived safety deficiencies in that early model reactor. The original date specified for closing them down was 2000, though subsequently 2006 and 2008 were agreed in relation to EU accession[c].

The latter dates were set despite major refurbishment being carried out on the units, including replacement of the emergency core cooling systems and modernizing the instrumentation and control systems[d]. The two Bohunice V1 reactors were the first V-230 units outside the Soviet Union and had had more upgrading work on them than any other of their type, costing some US$ 300 million since 1991[e]. An IAEA mission in 2000 reported that, for the V1 units, "all safety issues identified earlier by the IAEA have been appropriately addressed." Slovakia claimed that all their design safety deficiencies had been removed by the safety upgrading, and this had been confirmed by international expert safety review missions.

In the lead up to EU accession in 2004, nuclear industry representatives from Eastern Europe called for the introduction of transparent and rational EU safety standards rather than punitive closures of reactors which had been substantially upgraded. In particular, the Slovak Bohunice V1 units were cited as prime examples of the high safety standards which such reactors could achieve through upgrading with input from Western firms such as Siemens.

The units were producing electricity at half the average cost for all Slovak sources, and their closure before Mochovce 3&4 are on line has left the country short of power. Unit 1 of the Bohunice V1 plant was closed at the end of December 2006, eliminating about 9% of Slovakia's electricity supply. The second unit of the V1 plant was closed at the end of 2008. The Prime Minister said that he respected the decision to shut down the plant, but considered it as "energy treason" by the previous government, with Slovakia becoming an electricity importer. He suggested that it might be possible to restart the V1 units in future[f].

In 2013 the EU Parliament Committee on Budgetary Control complained that it was “unacceptable” that these reactors were not in irreversible shutdown.

From 2005 to 2008 operator Slovenské Elektrárne (SE) carried out a major modernization programme on the two Bohunice V2 units, to improve seismic resistance, cooling systems, and instrumentation and control (I&C) systems with a view to extending operational lifetime to 40 years (to 2025). This was followed by a progressive uprating programme of both units[1], eventually bringing the capacity of each unit from 440 MWe gross to 505 MWe gross (472 MWe net) by November 2010. The total cost was reported to be €500 million. SE is planning to extend the licences of the V2 units to 2045 following upgrading.

The state-owned Nuclear Decommissioning Company, Javys, owns Bohunice A1 and V1.

Mochovce

In 1982 construction on the first two units of the four-unit Mochovce nuclear power plant was commenced by Skoda, using VVER-440 V-213 reactor units. Units 1&2 started up in 1998 and 1999. These two units have been significantly upgraded and the I&C systems replaced with assistance from Western companies[9]. Uprates of 7% at Mochovce 1&2 were implemented by June 2008, taking them from 405 to 436 MWe net. Further uprates of turbines are under way to take them to about 466 MWe net (500 MWe gross) each by late 2021.

Mochovce reactor hall (Image: Slovenské Elektrárne)

Work on units 3&4 started in 1986 and halted in 1992. Construction was reactivated in mid-2009 and the units were expected to commence operation in 2012 and 2013. €2.8 billion was allocated to the completion project. See section below.

Slovenské Elektrárne

Bohunice V2 and Mochovce are owned and operated by Slovenské Elektrárne (SE). SE is owned 34% by the state through the National Property Fund, and 66% by Slovak Power Holding BV (SPH).

In 2006 Italian utility ENEL acquired a 66% stake in SE. In 2015 ENEL sought bids for its stake in SE. In June 2015 China National Nuclear Corporation (CNNC) said it planned to bid for ENEL's share of SE, but this did not proceed. In July ENEL said it would sell SE in two stages: first to reduce the 66% to below 50% in 2015, probably by selling to the state, then the rest once Mochovce 3&4 were completed. In December 2015 it was reported that ENEL had agreed to the sale of its SE equity to Czech-based energy company Energeticky a Prumyslovy Holdings (EPH), in two stages: initially 33% then the balance on completion of the Mochovce project. In July 2016 EPH closed the first phase of buying into SE through EPH's subsidiary, EP Slovakia BV. EP Slovakia became a 50% shareholder in SPH, with the other 50% owned by ENEL. Of the

total price of €375 million, €150 was paid immediately and the rest is due with the second phase of the transaction, a year after Mochovce 3&4 start operating. The second phase, involving the remaining equity of SPH owned by ENEL, will be another €375 million, subject to some adjustments.

New nuclear capacity

In October 2004 the government approved Italian ENEL's bid to acquire 66% of Slovenské Elektrárne for €840 million as part of its privatization process. Enel's subsequent investment plan approved in 2005 involved €1.88 billion investment to increase generating capacity, including €1.6 billion for completion of Mochovce units 3&4. The estimated cost of this project has more than doubled since then.

In January 2006 the government approved a new energy strategy incorporating these plans, which included capacity uprates at Mochovce 1&2 and the Bohunice V2 units.

Reactors under construction in Slovakia

Reactor Name	Model	Reactor Type	Gross Capacity	Construction Start
Mochovce 4	VVER V-213	PWR	471	1987-01-27

Plans for new nuclear build were outlined in the October 2008 Energy Security Strategy of the Slovak Republic, which incorporates the nuclear power plans in the 2006 Energy Policy. The 2008 Energy Security Strategy aimed to maintain the proportion of electricity generated by nuclear power plants at around 50% through the following measures:

- Completion of Mochovce 3&4 by 2013 (880 MWe gross).
- Uprates at Bohunice V2 and Mochovce 1&2 by 2010 (180 MWe gross).
- Uprates at Mochovce 3&4 by 2015 (60 MWe gross).
- Construction of a new reactor block at Bohunice by 2025 (1200 MWe gross).

In addition, around 2025, when the two V2 units will have reached 40 years of operation, the strategy called for either life extension of

the V2 units or for the construction of 1200 MWe of new nuclear capacity at Kecerovce in the east of the country[i].

The 2019 Integrated National Energy and Climate Plan for 2021 to 2030 stated that, on the basis of demand forecasts and transmission system requirements, the proposed 1200 MWe reactor at Bohunice would not be commissioned until after 2035. See below.

Mochovce 3&4

In February 2007, SE announced that it would proceed with Mochovce 3&4 completion and in July 2008, the European Commission approved the completion of the units subject to the design being brought into line with existing best practice for resistance to aircraft impacts[j,2]. Site works began in November 2008 and main construction activities commenced after contracts were signed in June 2009. A contract with Skoda JS, Russia's AtomStroyExport (ASE) and Slovak suppliers Výskumný Ústav Jadrovej Energetiky (VÚJE), Enseco and Inžinierske Stavby Košice was signed for more than €370 million to supply the remaining nuclear island equipment (beyond that delivered 20 years earlier), with part of the instrumentation and control (I&C) systems being from Siemens. Contracts for engineering, construction and project management of the conventional island were signed with Enel Ingegneria & Innovazione, and involve the use of Skoda Power steam turbines[3]. Following delay due to EU stress tests, startup was planned for 2013 and 2014. Uprating was planned by 2016, adding 62 MWe total to the original 880 MWe gross. By 2015, 170 major design changes had been implemented.

With the project depending substantially on the original 1986 construction permit including environmental clearance, it was challenged, and the need for a full new environmental impact assessment under European Union law was asserted[4].

In December 2012 Enel sought an extra €800 million on top of the original €2.8 billion estimate, and an extension of 22 months to complete the two units. In 2013, disagreement between ENEL and other shareholders continued, and the schedule was delayed. In August 2013 the government and Enel agreed to release €260 million to pay suppliers, and in April 2014 the government approved a revised budget of €3.8 billion for the plant, up from €3.25 billion agreed in August 2013. Then in June 2014 the regulator said that SE's schedule suggested that start-up would be delayed at least until mid-2015.

In October 2014 unit 3 was 80% complete and unit 4 60%. The design has evolved substantially from the original V-213. In October 2016 the estimated cost had grown to €5.4 billion. In March 2017 SE approved funding to complete the units, with operation then expected late in 2018 and late 2019. They were 95% and 83% complete then.

In May 2016 Rusatom Service signed an agreement with Czech Skoda JS to provide technical support for installation of primary circuit equipment, as well as for work on commissioning the two units.

In July 2018, Slovakia's prime minister stated that the new target dates for completion of units 3&4 were 2019 and 2020. However in April 2019, it was announced that a further delay of eight months was likely as a result of challenges to the permits needed for commissioning. Unit 3 was then 99% complete; unit 4, 85%. In April 2020 it was announced that unit 3 would not be operational until November or December 2020.

In May 2020 Slovakia’s Nuclear Regulatory Authority (UJDSR) announced that during post-installation checks it had found that material used in some pipe connections "did not meet specifications". As a result, SE is to carry out checks on "several thousand" components from the same supplier. UJDSR stated that it was not possible to determine the impact of the component checks on the timetable for Mochovce 3 at that time.

In February 2021 it was reported that opposition from Austrian environmental organization Global 2000 could delay fuel loading to the third quarter of 2021.

In May 2021 the regulator issued an operating licence for unit 3. In October 2021 SE said it expected final regulatory approval before the end of the year, allowing first electricity from unit 3 in early 2022.

In August 2022 the regulator issued the final authorization for Mochovce 3 to begin operating. SE began fuel loading on 9 September, and the unit achieved first criticality on 22 October.

Due to a small leak during pressure tests in December 2022, the startup schedule for Mochovce 3 was delayed by a few weeks. On 31 January 2023 Mochovce 3 was connected to the grid for the first time. The unit began commercial operation in October 2023.

In December 2024 SE began cold testing unit 4, and in March 2025 hot testing began.

In January 2025 Slovenské elektrárne outlined a district heating scheme to use waste heat from the Mochovce nuclear power plant to supply the town of Tlmače with implementation scheduled for 2027-2028.

Bohunice new block

The plans for a new reactor at Bohunice were announced in April 2008 for 1000-1600 MWe, probably using Western technology to enable MOX use. In December 2008, Czech utility CEZ was announced as the 49% joint venture partner, with state-owned Javys holding 51%. CEZ reportedly paid €117 million for its share. The formal joint venture JESS (Jadrová energetická spoločnosť Slovenska, Slovakia Nuclear Energy Company) agreement was signed in May 2009. Financing was to be finalized in 2011 and construction was planned to start in 2013, the expected cost then being €3.32 billion (for a 1200 MWe unit). Following an 18-month feasibility study there was to be a call for tenders, Areva and Westinghouse being considered the main possibilities, and in September 2012 JESS reported that the technology offered by six vendors* met the requirements for one or two reactors at Bohunice.

* Six vendors submitted information packages in December 2010: Westinghouse AP1000, Atmea 1100, Mitsubishi APWR 1700, Atomstroyexport MIR 1200, KHNP APR 1400 and Areva EPR 1600.

Earlier in August 2010 the newly-elected centre-right government said it was keen for the Bohunice project to proceed but would not offer any financial support for it. It was not expected to be operational before 2025, but the Minister of Economy said in May 2011 that it could be operational by 2020. A new left-wing government in April 2012 pledged to speed up the project and to decide upon proceeding early in 2013, as well as pushing completion of Mochovce 3&4, apparently delayed by the need to incorporate modifications due to EU 'stress tests'. The November 2014 energy plan affirmed the project.

CEZ sought to sell its 49% stake in JESS, reportedly for €110 million, in order to concentrate on its Temelin project, and Rosatom had been exploring the prospect of being both technology provider and investor in the new Bohunice unit. In April 2013 CEZ offered its JESS share to Rosatom, and the Slovak Economy Ministry said that it would be happy to accept this change, with Rusatom Overseas buying it and building a 1200 MWe reactor from about 2021. It sought a guaranteed long-term electricity price of €60-70/MWh, which the Economy Ministry refused to provide, and possibly a BOO

(build-own-operate) arrangement. Rosatom in January 2014 said it remained interested if the Slovak government could guarantee profitability in some way, and negotiations with CEZ continued. The government in December 2015 said Rosatom was still interested in the project, but Rosatom did not confirm this.

JESS arranged for the three-year environmental impact assessment [EIA] to be undertaken by AMEC and in April 2016 the Ministry of Environment announced its approval in principle for the project. JESS said then that about half of the preparations for the new reactor were complete and that it was working on criteria for the selection of a nuclear technology supplier.

Earlier in November 2015 a nuclear cooperation agreement was signed with China, focused on the fuel cycle supply chain and involving CNNC and CAEA. CNNC said it followed similar agreements with France and the UK.

The 2019 Integrated National Energy and Climate Plan for 2021 to 2030 stated that, on the basis of demand forecasts and transmission system requirements, the proposed 1200 MWe reactor at Bohunice would not be commissioned until after 2035. It said: "From the point of view of the overall balance of the Slovak Republic regulatory area... the construction of the new reactor... would be more effective if it was considered a substitute for the existing [Bohunice plant]. The simultaneous operation of EBO V2 and the New Nuclear Source would place additional demands on the electricity system of the Slovak Republic... due to the large accumulation of installed power in practically a single point of the transmission system."

In March 2022 CEZ said the project was effectively stalled. In February 2023 Jess applied to UJDSR for a siting permit for unit 5 at Bohunice.

In July 2023 JAVYS signed agreements with Westinghouse on the potential deployment of its AP1000 and AP300.

In June 2025 reactor developer Newcleo and JAVYS formally established a joint venture company, the Centre for Development of Spent Nuclear Fuel Utilisation (CVP), to develop a project to build up to four LFR-AS-200 lead-cooled fast reactors at Bohunice.

In August 2025 the prime minister confirmed a planned 1250 MWe Westinghouse unit for Bohunice. The same month, Slovenské elektrárne and Synthos Green Energy signed a memorandum of understanding to explore the deployment of GE Vernova Hitachi's BWRX-300 small modular reactors.

In October 2025 the prime minister announced his government had approved a proposed intergovernmental agreement with the USA on the construction of a new state-owned unit of over 1000 MWe at Bohunice. In January 2026, prime minister Robert Fico and US energy secretary Chris Wright signed the intergovernmental agreement in Washington. In January 2026 JAVYS selected Rothschild & Co as financial advisor for the new nuclear project after a competitive tender, with a contract valued at €13.9 million.

Reactors proposed in Slovakia

Site	Technology	MWe gross	Planned commercial operation
Bohunice New Block	AP1000?	1200	
Total (1)		1200	

Small modular reactors

In early 2024 Sargent & Lundy visited the Bohunice and Mochovce nuclear power plants and the Nováky and Vojany coal-fired plants to carry out initial site surveys for a feasibility study on the construction of SMRs in Slovakia. The work was carried out under Project Phoenix, a US initiative announced at the COP27 meeting that aims to support energy security and climate goals by creating pathways for coal-to-SMR power plant conversions.

In October 2024 Slovakia was awarded a $5 million grant from the USA's Nuclear Expediting the Energy Transition (NEXT) project. The money is to be used to help in the selection of a site for the construction of SMRs. In January 2026 a Project Phoenix feasibility study confirmed that four sites (Bohunice, Mochovce, Vojany and the U.S. Steel Košice site) are suitable for SMR deployment. Slovenske elektrarne has indicated SMRs could be operational in Slovakia from 2035.

Fuel cycle

Currently, all fuel supply is contracted from TVEL in Russia.

Assessing the viability of uranium mining in Slovakia was one of the priorities of the country's 2008 Energy Security Strategy. However,

in May 2014 the government resolved to ban uranium mining in the country unless it is approved by a referendum of local inhabitants, apparently due to opposition led by the mayor of Košice. The Slovak Environment Ministry proposed the amendment to the law, which came into effect in June 2015.

In May 2014 Perth-based Forte Energy agreed to take over the Slovakian uranium assets of European Uranium Resources (EUU, now trading as Azarga Metals Corp, formerly Tournigan Energy) based in Canada, through a share issue. This was not approved, so Forte agreed to pay $4 million in cash and farm-in for a 50% share in EUU subsidiaries Ludovika Energy and Ludovika Mining, which operate in Slovakia. The cash was paid in October 2014, and Forte then needed to spend $350,000 per year over ten years to complete the deal. However, in October 2015 Forte terminated the agreement and forfeited its 50% interest in the two companies to EUU due to the government failure to renew exploration licences.

EUU was investigating the Kuriskova and Novoveska Huta uranium deposits[k] as well as other uranium exploration targets in the Slovak Carpathian uranium belt in the east of the country. Kuriskova is about 15 km northwest of Kosice, and Novoveska Huta is about 50 km west of it, near Spisska Nova Ves. Both have potential molybdenum and rare earths by-products.

EUU in 2012 signed an agreement in principle with the Ministry of Economy for mining development to proceed. In April 2015 Ludovika Energy applied to extend the exploration licence for Kuriskova, but the Environment Ministry denied this. Ludovika Energy then applied for rare earths exploration licences over the Kuriskova deposit. EUU and Ludovika undertook legal action against the Environment Ministry regarding non-extension of the uranium permits, and expected compensation on the basis that more than €25 million had been invested in the project. EUU then agreed with a Slovak party to take over all operations, retaining a 25% stake in any financial compensation from legal proceedings. A regional court dismissed the lawsuit in June 2016. In May 2016 EUU changed its name to Azarga Metals and focused on a Russian copper-silver project.

In January 2015 Forte announced an updated resource estimate (JORC-compliant) for Kuriskova of over 12,900 tonnes U_3O_8 grading 0.525% U_3O_8 indicated resources, and inferred resources of 6140 tonnes U_3O_8 grading 0.153% U_3O_8. An underground mine using conventional techniques and with a cut-off of 0.05% U was earlier planned at a cost of US$ 225 million. Production of 530 tU/yr was envisaged from 2020.

The Novoveska Huta uranium deposit has 3680 t U_3O_8 measured and indicated resources at 0.064% U_3O_8 and 5900 t inferred resources (JORC-compliant) and a mining licence. There is potential for developing it with Kuriskova.

Following the war in Ukraine, nuclear power operators in EU countries that had previously relied on Russian-supplied fuel have sought alternative suppliers. In June 2023 Slovenské elektrárne signed a memorandum of understanding with France's Framatome to work together on the development of a 100% European nuclear fuel for VVER-440 reactors.

In July 2025 Slovenské elektrárne signed a contract with Urenco for the supply of enriched uranium to the Bohunice and Mochovce nuclear power plants until the mid-2030s to diversify its supplies.

In September 2025 Cameco signed a long-term agreement to supply uranium and conversion services to Slovenské elektrárne until 2036.

Radioactive waste management

Originally the policy was for used fuel to be disposed of without reprocessing, but in 2008 this changed to recycling it domestically.

At the beginning of 1996, the VYZ subsidiary of Slovenské Elektrárne was established for decommissioning nuclear facilities, radioactive waste and used fuel management. A separate subsidiary of Slovenské Elektrárne – Decom – was set up as a consultancy to focus on decommissioning. During ENEL's 2006 acquisition of a 66% stake in SE, the SE-VYZ subsidiary, along with the Bohunice V1 reactors (which were in operation at the time), was transferred to the state as the Nuclear Decommissioning Company (Javys)[1].

A treatment and conditioning plant for low- and intermediate-level wastes is operated by Javys at Bohunice, and a near-surface repository (the National Radioactive Waste Repository) at Mochovce began operation in 2001.

An interim wet storage facility for used fuel at Bohunice supplements reactor storage ponds, and has a capacity of 1680 tonnes (14,000 fuel assemblies). It has functioned since 1986 and is operated by Javys. Some used fuel was earlier exported to Russia for reprocessing (with Russia keeping the products).

Site selection for an underground high-level waste repository has commenced, although the country is also considering the option of participating in a shared international repository project[m].

In 2006 Slovakia set up the National Nuclear Fund to finance the costs of waste management and decommissioning of nuclear facilities. In 2018 the government proposed changes that will increase payment requirements for the country's sole nuclear operator, Slovenske Elektarne, as well as widening the scope of contributors to include nuclear facilities that do not generate electricity. The revised payment structure will see charges fixed for each facility instead of being based on installed capacity and a percentage of earnings. Since 2011 electricity consumers have been required to pay a nuclear levy, dependent on their electricity consumption.

“Due to high electricity prices” in 2021 the government suspended payments into the National Nuclear Fund for 2022 and 2023. Funds totalled €1853 million at the end of 2020, with payments that year of €75 million from the four operating plants, €37 million from Mochovce 3&4 under construction and €69 million from transmission companies. About €66 million was spent from the fund.

In December 2022 the government increased the annual overall payments from nuclear power plant operators to the National Nuclear Fund to €107.9 million in 2023, from €65.9 million in 2022.

Decommissioning

Decommissioning of the damaged Bohunice A1 reactor is continuing. Spent fuel was returned to Russia by 1990, leaving the reactor core and cooling contaminated with radionuclides from three fuel assemblies in the 1977 accident. Decommissioning is financed by the National Nuclear Fund.

Preparation for decommissioning the two Bohunice V1 reactors started in 2012, and the reactors were fully dismantled by the end of July 2022. Remaining equipment and systems at the site will be taken apart and processed by 2025, with power plant buildings then demolished with the site ready for redevelopment by 2027.

The Bohunice International Decommissioning Support Fund (BIDSF), administered by the European Bank of Reconstruction and Development (EBRD), was set up in 2001 to support the decommissioning of the Bohunice V1 plant, as well as to support

energy projects to help minimize the impact of the early closure of the reactors[n]. At the end of 2013 €225 million was made available by the EU via BIDSF, with support continuing to 2020. The National Nuclear Fund is co-financing the decommissioning of the V1 reactors.

In July 2025 the EBRD announced that the removal and dismantling of the Bohunice V1 reactors was completed.

Research and development

In May 2016 Rusatom Service signed an agreement with Slovak nuclear power plant research institute VUJE "to explore the possibilities of business cooperation in the field of maintenance and repair, operation support, training, supply of equipment and spare parts for nuclear power plants ... and the modernization of nuclear power plants around the world."

Regulation, safety & non-proliferation

The Nuclear Regulatory Authority of the Slovak Republic (UJDSR) is the independent regulatory body responsible for licensing, safety, waste management, radiation protection and safeguards.

Non-proliferation

The Slovak Republic is a party to the Nuclear Non-Proliferation Treaty (NPT) since 1993 as a non-nuclear weapons state. The Additional Protocol in relation to its safeguards agreements with the International Atomic Energy Agency was signed in 1999. The country is member of the Nuclear Suppliers Group, and since 2004, of Euratom.

Notes & References

Notes

a. Electricity consumption in 2019 was 4600 kWh per capita, up modestly from about 4400 kWh per capita in 1990. [Back]

b. Bohunice A1 was a channel-type KS-150 reactor with natural uranium fuel, heavy water moderator and carbon dioxide coolant (HWGCR) of 143 MWe gross which could be refuelled during operation. It was commissioned in 1972 and ran with many unplanned outages, including a significant accident in January 1976, until February 1977 when an accident rated Level 4 on the International Nuclear Event Scale led to its closure. Silca gel, used as a humidity absorber, was left covering a fuel assembly during refuelling operations, resulting in core damage and fission product contamination of the primary and secondary circuits. [Back]

c. In September 1999, the Slovak government adopted Resolution No. 801/99 to shut down the two Bohunice V1 units in 2006 and 2008. An earlier shutdown date of 2000 had previously been approved in September 1997 by the Slovak government in Resolution 684/97. The adoption of Resolution 801/99 was a condition of *Protocol No 9 on unit 1 and unit 2 of the Bohunice V1 nuclear power plant in Slovakia* of the Accession Agreement of the Slovak Republic to the European Union. [Back]

d. In 1996, the nuclear division of Siemens' Power Generation Group (KWU) – which, in 2000, became part of Framatome ANP (and later Areva) – was awarded a contract for the 'gradual reconstruction of Bohunice V1', which included the installation of digital instrumentation and control systems. The REKON consortium (Siemens with Slovak engineering company *Výskumný Ústav Jadrovej Energetiky*, VUJE) performed the design and implementation. The modifications were mainly carried out during the outages of the units from 1998-2000. This project demonstrated that the V-230 model of VVER-440 reactors could be upgraded to international safety standards. [Back]

e. Phase 1 upgrading of Bohnuice V1 was undertaken 1991-95, and phase 2 – intended to achieve Western European standards – through to 2000. In 2001, Slovakia relicensed Bohunice V1 units for another decade (until the next full safety review), though following the upgrades their operating lifetimes were expected to run until 2015. [Back]

f. The completion of Mochovce 1&2 included an extensive programme of design safety improvements involving Framatome, Siemens, and Electricité de France. The project started in 1996 and was completed in 1998 (unit 1) and 1999 (unit 2), the years in which the units began operation. [Back]

g. Early in 2009, only a few days after the shutdown of Bohunice V2-2 at the end of 2008, a gas dispute between Russia and

Ukraine affected nearly all of Slovakia's gas supplies. The Slovak government announced it had taken the decision to restart the reactor, but eventually the unit was not restarted. [Back]

h. Jadrová energetická spoločnosť Slovenska a.s. (Jess) is a joint venture between Slovak state-owned decommissioning company Javys (51%) and Czech utility CEZ (49%) established in 2009 to construct a new nuclear unit at the Bohunice site. See Slovakian nuclear JV gets government blessing, World Nuclear News (10 December 2009) [Back]

i. Kecerovce is named in the 2008 Energy Security Strategy as the site for a proposed 1200 MWe nuclear plant costing €3870 million. The plant was mainly intended to replace the nuclear capacity that will be lost when the Bohunice V2 units are shut down in 2025, depending on whether the 40-year operational lifetime of Bohunice V2 is extended. Although Kecerovce would be a new site, it had previously been considered as a potential site for a nuclear plant when Czechoslovakia was under Communist rule. [Back]

j. The European Commission (EC) said that completion of Mochovce 3&4 "fulfils the objectives of the Euratom Treaty," provided that a scenario including impact from an external source (such as a small aircraft) was developed and the necessary changes to withstand such an impact were implemented. In addition, the EC said it "remains the sole responsibility of the investor to ensure that the chosen design will provide an equivalent level of protection as a 'full containment'." It is not clear what this means, though the EC approval was simply to enable financing. The reactor's containment walls will be 1.5 metres thick and apparently the plant will be in compliance with safety requirements for existing reactors established in 2007 by the Western European Nuclear Regulators Association. [Back]

k. The Novoveska Huta uranium deposit, located approximately 65 km west of Kuriskova, was a historical uranium producer (closed in the early 1990s). Drilling completed between 1975 and 1985 reported a resource of 9000 tonnes U_3O_8 grading 0.075% U_3O_8. [Back]

l. Prior to Enel acquiring its 66% stake in Slovenské Elektrárne (SE) in April 2006, the Bohunice V1 reactors along with the decommissioning and waste management operations (under the SE-VYZ subsidiary) were transferred to the GovCo subsidiary. On Enel's acquisition of its stake in SE, GovCo was transferred to the government and later renamed as Javys (*Jadrová vyraďovacia spoločnosť*, Nuclear Decommissioning Company). [Back]

m. The Decom consultancy, with the Switzerland-based Association for Regional & International Underground Storage (ARIUS), ran a European Union-funded project to undertake a pilot study on the technical and legal requirements for a regional waste repository. This SAPIERR (Support Action: Pilot Initiative for European Regional Repositories) project ran from the end of 2003 for two years, and was followed by SAPIERR II, which finished in January 2009. The main proposal from the SAPIERR project was to set up a European Repository Development Organisation (ERDO). An ERDO Working Group is currently developing a consensus model for ERDO. Eight of the 14 countries that participated in the SAPIERR project are members of the ERDO Working Group (Bulgaria, Italy, Lithuania, Netherlands, Poland, Romania, Slovakia and Slovenia). See page on International Nuclear Waste Disposal Concepts [Back]

n. A webpage on the Bohunice International Decommissioning Support Fund is on the EBRD website (www.ebrd.com). [Back]

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